

Molcano: Molecular Language for Chemical Assembly Notation

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MOLCANO: MOlecular Language for Chemical Assembly NOtation

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Abstract

Molecular design operates through fragment-based reasoning, yet many widely used molecular encodings rely on atom-level representations. This misalignment makes fragment-by-fragment construction and scaffold-centered designs hard to realize, especially for fused-ring systems. We present MOLCANO, a molecular language that represents molecules as chemically interpretable fragments linked by explicit junction tags, supporting single-bond, face-to-face, and even substructure-level connections for versatile scaffold assembly. It facilitates the construction of chemically complex structures while maintaining interpretability throughout the generation process. We validate MOLCANO across three open datasets spanning small-molecule drugs and organic electronic materials, covering de novo generation, superstructure growth/decoration, and medicinal-chemistry scaffolding. This also integrates with reinforcement learning for goal-oriented optimization of properties and with large language models for interactive, text-guided editing. By aligning computational representations with chemists' modular reasoning, MOLCANO broadens the accessible chemical design space and accelerates molecular innovation in drug discovery, materials science, and beyond.

Keywords: Molecular Representation, Fragment-Based Representation, Generative Model, Materials Discovery

1 Introduction

The language we use shapes the contours of our thought. Our patterns of thought should be reflected in the languages we construct. Limited expressiveness narrows our ability to understand, to imagine, and to create. The chemical language that we use to encode molecular structures formulates how artificial intelligence perceive, speculate, and design. Existing chemical languages commonly adopted in molecular representation, however, do not reflect how chemists actually conceive molecules, leading to a misalignment between computational encoding and human chemical intuition.

The misalignment becomes apparent in standard laboratory practices, where chemists typically start with a seed scaffold and then enlarge the molecule fragment by fragment—via ring fusion, linker insertion, or substituent grafting. Each step is guided by criteria such as ready availability, low cost, supportive structure–activity relationships (SAR) or structure–property relationships (SPR) data, and overall synthetic tractability (e.g., accessible reagents and robust reaction conditions). By contrast, string notations like SMILES order atoms according to a depth-first traversal of the molecular graph; the atoms belonging to a single fragment (e.g., an entire fused ring or pendant group) are therefore dispersed throughout the sequence[1]. For an AI model taught to use this language, identifying or altering a fragment becomes a non-local, syntactically fragile read/edit rather than the single, coherent step a chemist envisions. In effect, chemical AI and bench chemists are *speaking different dialects*: the former manipulates atom tokens, while the latter thinks in modular building blocks—making the journey from in-silico concept to physically realized molecule slower and more error-prone than necessary.

Several efforts have sought to close this human-machine gap by adopting fragment-based molecular representations. Typically, these methods decompose molecules into fragments—via algorithms such as BRICS[2], MMPA[3], or RECAP[4]—and then encode each fragment as SMILES and its variants, or entirely new string representations[5–10]. However, these approaches yield incomplete fragmentation: since the algorithms only process acyclic single bonds, fused rings and other cyclic substructures remain intact as single fragments. The result is coarse modularity that inherits core liabilities of SMILES—non-local fragment edits and syntactic brittleness. This incompleteness also limits design diversity, e.g. scaffold elaboration via ring fusion/annulation cannot be handled. This shortcoming is especially consequential in domains where aromatic frameworks control function, from tuning protein–ligand recognition or selectivity in medicinal chemistry[11] to tuning electronic/optical response in organic materials[12–14]. Overall, predictive and generative models combined with these representations can miss a large, chemically and pharmacologically rich portion of chemical space as well as chemical modularity and design flexibility.

To address these limitations, we introduce MOLCANO—*Molecular Language for Chemical Assembly Notations*—a flexible line notation that represents molecules as sequences of user-defined molecular fragments annotated with junction tags specifying unrestricted connections between them. Users can fully customize the fragment vocabulary—tailoring it to reagent availability, SAR/SPR insights, and intuitive reading and editing for molecular structures. The fragmentation includes not only to single-bond linkers but also to fused-ring systems into chemically meaningful subunits. MOLCANO’s

fragment-based encoding embeds chemists’ expert reasoning directly into the training data, enabling predictive or generative models to naturally learn and replicate domain expertise.

In the following sections, we first present MOLCANO notation using ν -DABNA skeleton as an illustrative example in Section 2.1. We then report the main results on superstructure generation in Section 2.2, followed by evaluations of *de novo* generation benchmarks in Section 2.3. Next, in Section 2.4, we demonstrate goal-oriented generation with MOLCANO representations, employing reinforcement learning to steer the model toward desired property distributions. We also describe our application of MOLCANO to a proprietary dataset of blue fluorescence emitters, which outlines real-world deployment scenarios. Key finding and future directions are summarized in Section 3. Detailed descriptions of the MOLCANO decomposition algorithm, data augmentation, datasets, and experimental setups are provided in Section 4.

2 Results

2.1 Molecular Language for Chemical Assembly Notations

A MOLCANO string is composed of multiple fragments separated by a dot ($.$), with every fragment being a valid SMILES string. Chemical assembly between fragments is specified by assigning integer ATOMMAPNUMBERS to designated atoms (e.g., [C:1]). Atoms with the same ATOMMAPNUMBER across different fragments are treated as the *identical* atom in the original molecule. The full molecular graph is recovered by overlapping all atoms that share an identical ATOMMAPNUMBER, ensuring exact reconstruction of the original structure.

DABNA-1, a rigid polycyclic aromatic emitter that serves as a core unit in various organic electronic applications, provides a representative example of how MOLCANO captures chemically meaningful fragment assembly. Its photophysical performance can be enhanced by fusing two DABNA-1 cores through a central benzene bridge, forming ν -DABNA skeleton^[15] as shown in Fig. 1. In MOLCANO notation, this assembly is transparently encoded as two DABNA-1 fragments—each a valid SMILES string denoted with explicit junction tag on the overlapping benzene atoms—separated by the delimiter.

Overlaps can occur at multiple scales: a single atom (e.g., substituent attachment as in red scissor of Fig. 2a), a bonded pair of neighboring atoms (edge-sharing in ring fusions as in blue scissor of Fig. 2a), an entire ring (as in ν -DABNA example, Fig. 1), or even larger polycyclic motifs. MOLCANO handles all of these by assigning the same ATOMMAPNUMBER to each atom in the overlap set, producing multi-atom junctions when required while preserving correct connectivity.

Users may choose fragments based on functional-group rules, common synthetic building blocks, bioactive scaffolds, or any other “flavor” of chemical intuition. With its fully user-configurable fragment vocabulary, MOLCANO allows researchers to define collections of chemically meaningful substructures tailored to their specific design goals.

At the current stage, domain expertise remains important for the effective use of MOLCANO, as the framework relies on user-defined fragment vocabularies to encode chemically meaningful inductive bias. As demonstrated in this work, fragment selection follows practical, domain-dependent heuristics that lead to different fragment distributions across applications. In drug discovery, fragments are chosen to preserve pharmacophoric cores while enabling optimization of properties such as solubility, metabolic stability, or binding affinity. In contrast, for organic electronic materials, where extended π -conjugation governs electronic and optical properties at the level of the entire molecule, both the core structure and attached fragments must support conjugation; accordingly, fragment vocabularies are intentionally biased toward conjugated and aromatic motifs. These practical heuristics ensure that generated superstructures remain chemically meaningful and relevant to their intended domains.

2.2 Superstructure Generation

Superstructure generation refers to the process of expanding given scaffolds with additional fragments. This approach is widely used in drug discovery and materials design to optimize molecular properties, such as solubility, metabolic stability, or binding affinity, while preserving the core functionality of the parent structure.

We evaluated MOLCANO’s applicability to drug analogue design using two representative scaffolds: camptothecin (CPT) and tetracycline. CPT is a natural anticancer alkaloid characterized by a rigid pentacyclic core framework^[16]. Clinically approved or investigational CPT derivatives—including topotecan, irinotecan, rubitecan, and exatecan—were developed through targeted structural modifications to improve CPT’s poor aqueous solubility and lactone instability and to enhance potency^[16–20]. While some analogues can be obtained through simple substituent changes, others require more complex edits—such as formation of an additional fused ring or simultaneous modification of multiple ring junctions—yielding structures that challenge conventional molecular representations. To demonstrate MOLCANO’s ability to represent and generate structurally diverse superstructures, we fine-tuned a pretrained GPT-style model to generate analogues conditioned on a partial CPT scaffold. As shown in Fig. 2, the model produced diverse analogues, including complex fused-ring architectures that are difficult to obtain with conventional approaches.

We also tested MOLCANO for tetracycline analogue generation using in-context learning (ICL). By prompting ChatGPT-4o with a concise syntax description and a few-shot examples—without any parameter updates—the model generated diverse tetracycline derivatives. However, it occasionally produced outputs that violated MOLCANO syntax or lacked chemical validity. Adding an explicit instruction emphasizing syntactic correctness and chemical realism significantly improved generation quality. Representative examples can be found in Supplementary Information Fig. S1 and the complete prompt used in this experiment are provided in Table S1.

To further demonstrate the applicability of MOLCANO under constrained design settings, we consider Securinega alkaloids as a representative case study involving a limited fragment library. This family of natural products, occurring in

species of the genera *Securinega* and *Phyllanthus*, encompasses structurally varied, nitrogen-containing polycyclic scaffolds with documented biological activities—including central nervous system stimulation, anticancer effects, neuroprotection, and immunomodulation[21]. Structurally, Securinega alkaloids share conserved core scaffolds but exhibit wide variation in ring sizes, ring-fusion patterns, and substituent replacements[22, 23]. We curated a compact fragment vocabulary from characterized Securinega derivatives; these fragments approximate biosynthetic building blocks, enabling MOLCANO to capture chemically meaningful assembly motifs observed in nature.

Fig. 3 illustrates an interactive workflow for fragment-constrained superstructure generation with MOLCANO. The user selects a core scaffold and constrains assembly to a predefined fragment library. With this fragment vocabulary supplied in the prompt, the model generates diverse structural variants that reflect known biosynthetic assembly patterns. Representative examples of the generated superstructures are presented in Supplementary Information Fig. S2.

2.3 De novo generation

Dataset	Metric	MOLCANO	SELFIES	SAFE	SMARTS	SMILES	FRAGSMILES	TSMILES
DRUG	Validity	0.9170	0.9996	0.7054	0.9148	0.9246	0.9576	0.9996
	Uniqueness	0.9915	0.9921	0.9863	0.9872	0.9859	0.9832	0.2140
	Novelty	0.8794	0.8453	0.7965	0.8006	0.7848	0.6069	1.0000
	Efficiency	0.7995	0.8383	0.5542	0.7230	0.7154	0.5714	0.2139
	Quality	0.4808	0.3492	0.3861	0.4165	0.4099	0.4193	0.2607
	LRE _{≥8} (%)	-10.68	399.5	122.9	5.346	45.01	505.6	-77.31
MOSES	Validity	0.9842	1.0000	0.9860	0.9259	0.9973	0.9978	0.9993
	Uniqueness	0.9994	0.9948	0.9795	0.9904	0.9878	0.9988	0.2734
	Novelty	0.9115	0.7830	0.7316	0.7449	0.7015	0.8108	1.0000
	Efficiency	0.8966	0.7789	0.7065	0.6830	0.808	0.8080	0.2732
	Quality	0.9692	0.9651	0.9685	0.9777	0.9694	0.9636	0.2647
OEMAT	Validity	0.9291	0.9993	0.6693	0.9307	0.9353	0.9269	0.9980
	Uniqueness	0.9982	0.9980	0.9882	0.9937	0.9931	0.9927	0.4041
	Novelty	0.9120	0.9174	0.7507	0.7357	0.7616	0.6879	1.0000
	Efficiency	0.8459	0.9149	0.4965	0.6803	0.7074	0.6329	0.4033
	LRE _{≥8} (%)	-57.11	1678.0	226.5	-38.85	-28.55	133.1	11.56

Table 1 Efficiency, quality, and structural invalidity measured as large-ring enrichment (LRE_{≥8}) in de novo generation. Quality metric is omitted for the OEMAT dataset due to uncertain relevance of drug-likeness. LRE values are omitted for the MOSES dataset because $p_{\text{train}}^{(\geq 8)} = 0$.

We conducted de novo generation experiments to compare MOLCANO with existing molecular representations. Decoder-only transformer models were trained on a large corpus of molecules using various string-based representations to learn general chemical patterns. For MOLCANO training, we applied the data augmentation techniques described in Section 4.2 to enhance the generation performance. We evaluated generation performance using three metrics—validity, uniqueness, and novelty—following

the protocol in [24], and we define generation efficiency as the fraction of generated molecules that satisfy all three criteria. We also adopt a quality metric, i.e., the fraction of generated molecules that are highly drug-like and synthetically accessible, as introduced in [25]. In addition, we report a structural invalidity measure, *large-ring enrichment* ($\text{LRE}_{\geq 8}$), which quantifies the relative overproduction of ≥ 8 -membered rings in the generated set versus the training corpus:

$$\text{LRE}_{\geq 8} = 100 \frac{p_{\text{gen}}^{(\geq 8)} - p_{\text{train}}^{(\geq 8)}}{p_{\text{train}}^{(\geq 8)}}\%, \quad (1)$$

where $p_{\text{gen}}^{(\geq 8)}$ and $p_{\text{train}}^{(\geq 8)}$ denote the fractions of rings with size ≥ 8 among all rings in the generated and training sets, respectively. Higher values indicate greater structural invalidity (0 indicates a match). By sampling 10K candidates across multiple runs, we computed the mean value of each metric for different molecular representations. All generated molecules are canonicalized at the molecular graph level (i.e., using canonical SMILES) prior to evaluation to ensure fair and reproducible comparison across models and representations. The results are summarized in Table 1. MOLCANO outperformed or matched all other representations across all domains. Although SELFIES achieved higher generation efficiency on DRUG and OEMAT, it came at the cost of structural validity, frequently producing chemically implausible large rings (≥ 8 -membered), which led to very high large-ring enrichment ($\text{LRE}_{\geq 8}$). Illustrative examples are provided in Supplementary Information Fig. S3. This underscores that conventional efficiency metrics, while useful as a baseline, should be complemented with measures of chemical plausibility (e.g., quality and large-ring enrichment, $\text{LRE}_{\geq 8}$).

2.4 Goal-oriented generation

Experiment	QED	SAS	Similarity	Efficiency
De novo generation	0.5762	3.1802	0.8091	0.742
Property optimization	0.8604	2.3406	0.7988	0.7558
Domain-adaptive generation	0.8378	2.3743	0.8317	0.7646

Table 2 Goal-oriented generation results on multiple properties, as well as the similarity toward a target domain dataset.

Goal-oriented generation in molecular discovery frames generation not as blind sampling from a learned distribution, but as an optimization problem: producing novel structures that maximize one or more desired criteria. We consider two settings: property optimization and domain-adaptive generation. Specifically, we optimize for quantitative estimation of drug-likeness (QED) and synthetic accessibility score (SAS). For domain-adaptive generation, the objective is to shift the generation distribution toward a target domain by maximizing similarity to target samples as well as the

properties. In Table 2, desired properties are optimized for both property optimization and domain-adaptive generation tasks, where the number of generated sample is 1K and the reward functions are described in Section 4.3. While synthetic accessibility is evaluated using SAS in these tasks, we note that SAS serves as a lightweight proxy for synthetic accessibility. In practical use, this step may be followed by more in-depth validation using additional retrosynthetic analysis tools. Detailed examples and representative synthetic routes are provided in the Supplementary Information Fig. S4.

Fig. 4a) illustrates the resulting distribution shifts of the domain-adaptive generation.

Precise control over the emission spectrum provides a representative real-world test case for such goal-oriented optimization, particularly in the design of blue fluorescence emitters. By adjusting the π -conjugation or fused ring connectivity, researchers can achieve peak shifts in photoluminescence. The MOLCANO representation offers the flexibility to easily amalgamate fused rings, which are challenging to handle with conventional molecular representations.

We tested the effectiveness of MOLCANO in a real-world molecular design task of generating blue fluorescence emitters with specific spectral and electronic properties. Although the details of the dataset remains proprietary, the task setup reflects a high-stakes industrial application. The goal is to discover a novel blue fluorescence emitter with desired peak wavelength of the photoluminescence spectrum (PL_{max}), full width at half maximum ($FWHM$), and the highest occupied molecular orbital energy ($HOMO$). Achieving all three objectives concurrently is challenging, as decreasing PL_{max} and $FWHM$ while increasing $HOMO$ often involves trade-offs in the underlying molecular structure. An AI model capable of exploring diverse candidate structures that satisfy these competing criteria can significantly aid the discovery process.

We trained a generative model on our internal dataset consisting of various emitter molecules and used a predictive model[26] as a proxy for property estimation. Fig. 4b) illustrates the distribution shift toward target property ranges. Despite the structural complexity of blue fluorescence emitters compared to typical drug molecules, MOLCANO-GPT successfully generated candidate molecules with the desired properties.

3 Discussion

We introduced MOLCANO, a fragment-centric molecular language that provides a flexible framework enabling the alignment of computational representations with fragment-based reasoning employed by chemists. The chief benefit of MOLCANO is to make fragment-level design first class. This improves editability, interpretability, and data efficiency, and expands the reachable chemical design space for early-stage drug discovery while remaining applicable to functional materials.

In generative modeling, MOLCANO enables fragment-by-fragment construction for both de novo molecular design and superstructure generation, including scaffold-preserving growth and decoration. The framework integrates naturally with large language models, allowing in-context learning with models like ChatGPT-4o without

additional training, and with reinforcement learning for property-driven optimization. We demonstrated concrete scenarios for applying the proposed syntax in drug discovery, such as generating hexacyclic camptothecin analogues and multi-fused Securinega alkaloids. Industrial applicability is further validated through real-world blue fluorescence emitter design.

More broadly, the fragment-centric formulation of MOLCANO is, in principle, applicable across molecular scales, from small molecules to very large systems, by defining appropriate fragment abstractions and connectivity rules. As molecular size increases, the resulting overhead from longer token sequences remains relatively modest, whereas the dominant challenge arises from the increasing complexity of fragment connectivity patterns, which typically requires greater data coverage to learn reliably.

Building on this observation, several directions remain. First, automatic discovery and adaptation of fragment vocabularies (data- and task-aware) could further reduce manual curation. Second, incorporating synthesis awareness (reaction rules, retrosynthetic feasibility, and cost/availability constraints) and richer stereochemical/3D information would strengthen real-world utility. One promising direction is to enrich fragment junctions with geometric cues to better reflect three-dimensional structure. Third, standardized benchmarks for superstructure generation and scaffold-centric tasks, coupled with human-in-the-loop evaluations, will better quantify practical impact. Finally, tighter integration with active learning and experimental feedback loops may accelerate iteration from in silico proposals to validated compounds and materials.

4 Methods

4.1 Formal Definition

Let G be a molecular graph with vertex set $V(G)$ and edge set $E(G)$. For a set of subgraphs $\mathcal{K} \subseteq \text{Subgraphs}(G)$, define the covered vertex and edge sets by

$$\bigcup_{h \in \mathcal{K}} V(h) \quad \text{and} \quad \bigcup_{h \in \mathcal{K}} E(h), \quad (2)$$

respectively. We write $G - \mathcal{K}$ for the subgraph G obtained by removing all vertices $\cup_{h \in \mathcal{K}} V(h)$ and edges $\cup_{h \in \mathcal{K}} E(h)$.

In practice, a convenient default strategy is to fragment each molecule into its the smallest set of smallest rings (SSSR) plus acyclic substituent groups. We first compute SSSR for the molecular graph and treat each ring in this basis as a fragment and then acyclic branches (functional groups, side chains, linkers) are extracted as substituent fragments attached to specific ring atoms. This SSSR+substituent factorization gives a compact, chemically interpretable scaffold backbone while isolating modifiable units for enumeration, scaffold hopping, and ring-fusion design. Algorithm 1 illustrates the detailed procedure for fragmentation a molecule with predefined structures as well as SSSR.

Algorithm 1 MOLCANO Fragmentation of a Molecule

Input: Molecular graph G , predefined fragment patterns $\mathcal{A}$
Output: SMILES fragments F with atom-map annotations on overlapping atoms

```

1:  $\mathcal{K} \leftarrow \{s \in \text{Subgraphs}(G) \mid \exists p \in \mathcal{A} : \text{matches}(s, p)\}$ 
2:  $\mathcal{L} \leftarrow \text{SSSR}(G - \mathcal{K})$  // smallest set of smallest rings
3:  $\mathcal{M} \leftarrow \text{AcyclicSubstructures}(G - (\mathcal{K} \cup \mathcal{L}))$ 
4:  $S \leftarrow \mathcal{K} \cup \mathcal{L} \cup \mathcal{M}$  // full fragment set
5:  $F \leftarrow \emptyset$ 
6: // identify overlapping atoms (present in  $\geq 2$  fragments)
7:  $O \leftarrow \{v \in V(G) \mid |\{f \in S : v \in V(f)\}| \geq 2\}$ 
8: initialize empty map  $\text{ID} : O \rightarrow \mathbb{N}$  // ATOMMAPNUMBERS
9: for all  $v \in O$  do
10:   if  $\text{ID}(v)$  undefined then
11:     assign a new integer  $\text{ID}(v)$ 
12:   end if
13:   write  $\text{ID}(v)$  onto every occurrence of  $v$  in each  $f \in S$  with  $v \in V(f)$ 
14: end for
15: for all  $f \in S$  do
16:    $s \leftarrow \text{FragmentToSmiles}(f)$ 
17:    $F \leftarrow F \cup \{s\}$ 
18: end for
19: return  $F$ 

```

4.2 Data Augmentation

By design, a MOLCANO string is permutation-invariant in two key respects. First, the order of fragments in a MOLCANO string is arbitrary—permuting fragment positions does not alter the molecular graph it represents. Because the number of possible fragment permutations is computationally intractable, providing all permutations to a model with finite capacity would introduce excessive redundancy. To mitigate this, we constrain fragment ordering using a graph-traversal procedure. First, we construct a fragment-level connectivity graph in which each node corresponds to a fragment and each edge represents a connection defined by shared ATOMMAPNUMBERS. From this starting point, we perform a traversal—such as depth-first search (DFS) or breadth-first search (BFS)—so that each subsequent fragment in the sequence is directly connected to at least one fragment already placed. This approach preserves structural connectivity while substantially reducing redundant permutations, maintaining the benefits of permutation invariance without overwhelming the model with equivalent encodings.

Second, ATOMMAPNUMBER identifiers can be reassigned to any non-conflicting integers without affecting the underlying molecular graph. In the MOLCANO representation, these identifiers serve only to indicate which atoms in different fragments correspond to the same atom in the reconstructed molecule; their numeric values themselves carry no intrinsic chemical meaning. Consequently, any renumbering that preserves these mapping relationships yields an equivalent structure. To generate such renumbered variants, we iterate over atoms with assigned map numbers, randomly

select another atom with a map number, and swap their values, ensuring that each mapping remains consistent across all fragments. This procedure preserves molecular connectivity and composition while introducing additional representational diversity.

These invariances ensure that multiple equivalent string representations correspond to the same molecule, while introducing beneficial variability into the training data. Such variability functions as a form of data augmentation for generative chemical models. By exposing the model to diverse but equivalent encodings of the same structure, permutation invariance reduces overfitting to a single canonical form, encourages learning of chemically relevant relationships rather than positional biases, and improves robustness when generating novel molecules. This effect is analogous to geometric transformations in image-based learning, where rotated or flipped images enhance generalization without changing the semantic content.

All experiments use a fixed BFS-based fragment ordering to ensure consistency across runs, while AtomMapNumber permutations are applied as controlled augmentation during training.

4.3 Experimental Setups

We conducted experiments on three benchmark datasets. The MOSES dataset contains 1.6M molecules and is widely adopted as a benchmark for molecular generation[27]. The DRUG dataset comprises 340k drug-like molecules curated from diverse sources, including natural products and steroids from PubChem, similarity searches involving exatecan, camptothecin, artemisinin, and cholestane[28], as well as the COCONUT natural product database[29]. The OEMAT dataset contains 382k organic electronic materials, collected from multiple sources, including chromophores[30], fluorescent molecules[31], QM40[32], high-throughput Iridium(III) complex data[33], organic photovoltaic (OPV) database[34], and optoelectronic dataset[35].

For each molecule, we prepared representations in SMILES[1], SMARTS, SELFIES[36], SAFE[9], and MOLCANO formats, and train Byte-Pair Encoding (BPE) tokenizers[37] and generative models for each format. After preparing the tokenizers, we trained transformer models to minimize the next token prediction loss, following the GPT-2 architecture. The best model was selected based on evaluation loss on a held-out dataset. For reproducibility, we utilized the Hugging Face Transformers[38] library. Key hyperparameters for training included learning rates, dropout rates, and random seeds for model initialization. For smaller datasets such as DRUG and OEMAT, higher dropout rates were applied to reduce overfitting.

For property optimization, the reward function is defined as a weighted sum of reward values for each property. Specifically, we use the reward function $R_{\text{prop}} = R_{\text{QED}} * 2.0 + R_{\text{SAS}} * 0.1$, where larger values of QED leads to higher rewards while smaller values of SAS results in higher rewards. R_{QED} is clipped for values > 0.8 , and R_{SAS} is clipped when < 2 . During training, the model generates new candidates, calculates rewards for each sample, and updates parameters to maximize the reward. For domain-adaptive generation, the reward functions used are

$R_{\text{sim}} = \text{mean}(\text{top-k similarities})$ and $R_{\text{prop+sim}} = R_{\text{prop}} + R_{\text{sim}}$, combining property optimization with domain alignment. Even with a small number of target samples, the model can be optimized to produce target-like molecules. We apply Group Relative Policy Optimization (GRPO)[39] to optimize pretrained models.

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Figure legends

Figure 1. ν -DABNA skeleton and its Molcano representation.

ν -DABNA skeleton is formed by linking two DABNA-1 units through a central benzene ring. MOLCANO encodes this process by treating each DABNA-1 as a fragment and assigning identical ATOMMAPNUMBERS to overlapping atoms. In contrast, conventional representations such as SMILES struggle to highlight the DABNA-1 fragments within the ν -DABNA structure, underscoring the difficulty of expressing chemically meaningful scaffolds.

Figure 2. Camptothecin (CPT) analogues design with Molcano.

- a) Scheme of CPT molecular graph to MOLCANO representation.
- b) Clinically investigated CPT derivatives.
- c) Superstructures generated using MOLCANO representation, illustrating its ability to capture complex scaffold modification.

Figure 3. Interactive superstructure generation with Molcano.

A user instructs the model to generate Securinega alkaloid derivatives by selecting a core scaffold and restricting the assembly to a predefined fragment library. The large

language model responds with a valid MOLCANO string. When merged, the output reconstructs the complete molecule, which can also be expressed as a standard SMILES. This dialogue illustrates how MOLCANO enables fragment-constrained, scaffold-based molecular design in a transparent and interpretable way.

Figure 4. Distribution shift for drug-like molecules and blue fluorescence emitters.

a) Comparison of molecules generated by the de novo model (blue) and the optimized model (orange) for drug-like molecules. The optimized model shifts the distributions toward higher quantitative estimate of drug-likeness, lower synthetic accessibility score, and greater similarity to the target domain.

b) Comparison of molecules generated by the de novo model (blue) and the optimized model (orange) for blue fluorescence emitter design. The optimized model produces a larger proportion of molecules within the target ranges of peak wavelength and highest occupied molecular orbital energy, while also achieving narrower full width at half maximum, indicating effective multi-objective optimization.

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Data availability

The MOSES dataset is publicly available at <https://github.com/molecularsets/moses>. Curated datasets for Drug-like molecules (DRUG) and Organic Electronic Materials (OEMAT) were compiled from open sources, with details provided in Section 4.3. The blue fluorescence emitters dataset analysed in this study contains confidential proprietary information and is not available.

Code availability

The source code associated with this study is publicly available at <https://github.com/samsungDS-PoCs/MOLCANO>.

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Hwidong Na, Eun Hyun Cho, and MiYoung Jang contributed equally to this work. The first author order was determined by a coin flip.

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Contributions

H.N, E.C. and M.J conceived the idea, performed the experiments, and wrote the manuscript. J.H, C.O., S.P, J.W, S.Y contributed to improving the idea. H.L., H.K., J.K., J.K, S.L, K.K. contributed to interpreting the results. All authors reviewed the manuscript.

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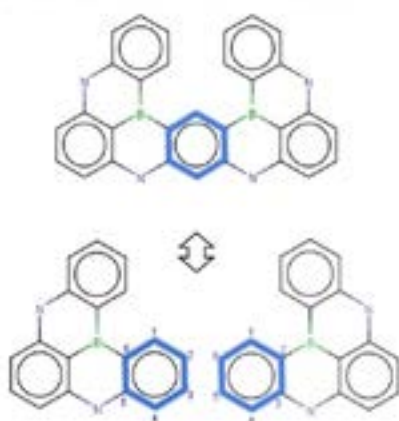
Correspondence to Hwidong Na, Eun Hyun Cho, MiYoung Jang.

Ethics declarations

Competing interests

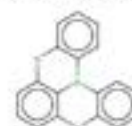
All authors declare no financial or non-financial competing interests.

🔍 v-DABNA skelton as Fused DABNA-1 units



✖ Customizable Fragment Library

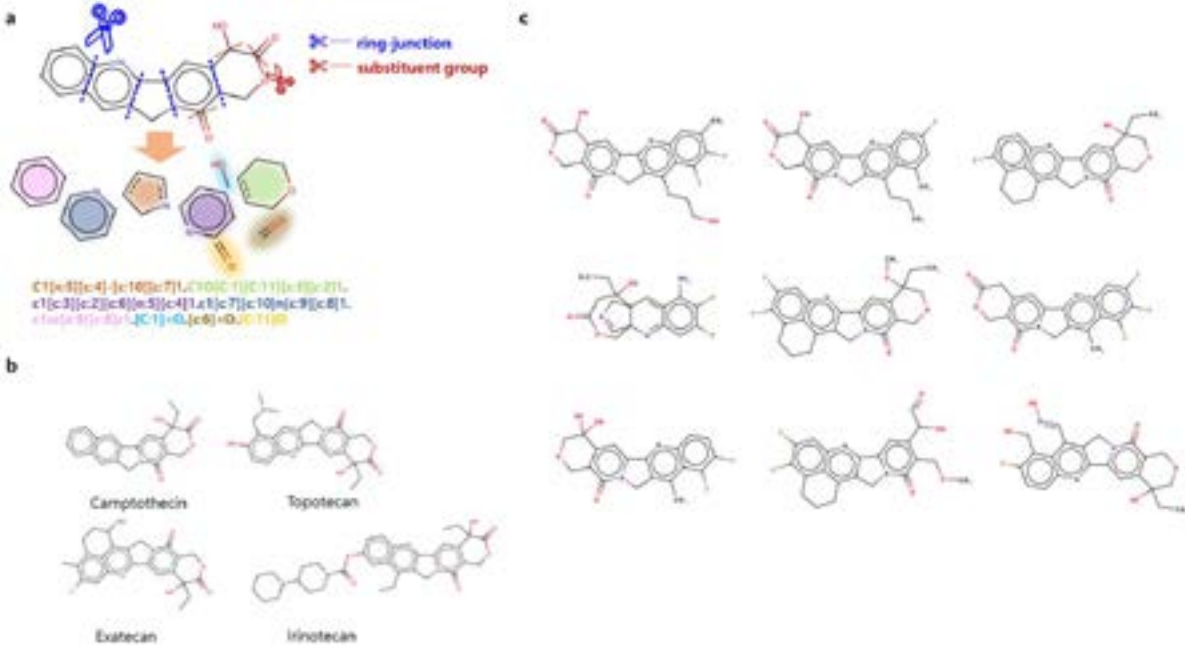
MOLCANO



N1c2ccccc2B2c3cc4B5c6ccccc6Nc6cccc(Nc4cc3Nc3cccc1c23)c56



N1[e:5]2[e:4][e:3][e:2][e:1][e:6]2B2c3ccccc3Nc3cccc1c23.
N1c2ccccc2B2[e:2]3[e:1][e:6][e:5][e:4][e:3]3Nc3cccc1c23

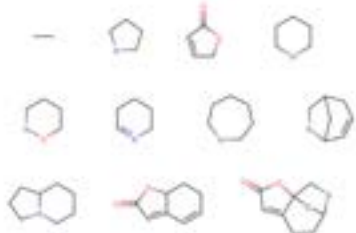


I want to generate securinega alkaloid derivatives. Given a core scaffold, please perform superstructure generation using the Molcano syntax with fragment-level assembly. The core scaffold should be chosen from the options shown below, and the available fragment library is also shown below.

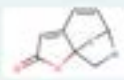
- Core Skeletons



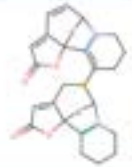
- Fragment Library



Understood. Please select a core scaffold.



Got it—when recombined, this corresponds to:



Type your message here

